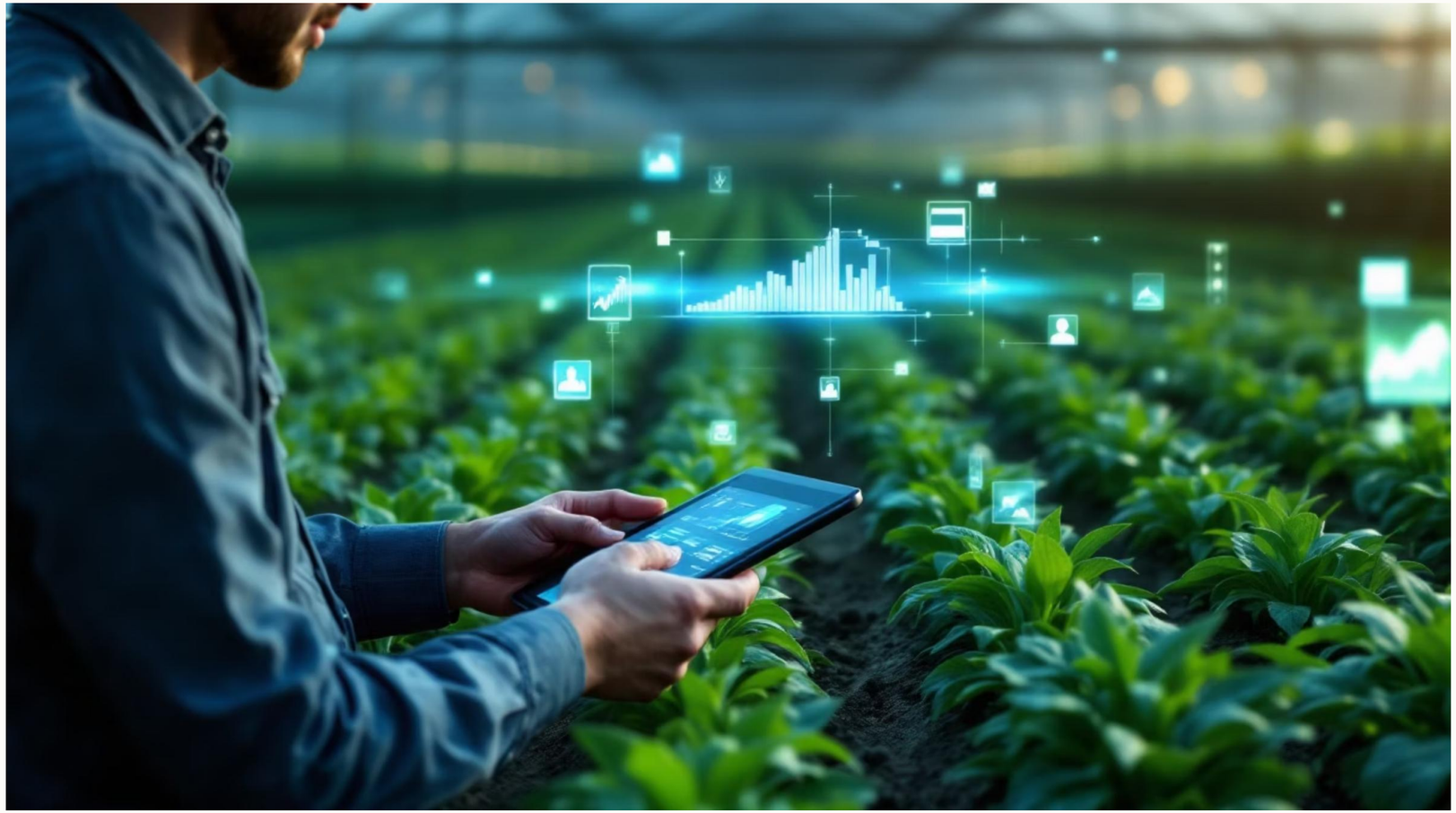


AgroSmart DZ

Data-Driven Agriculture for Algeria

Team Roles

Frontend Developer

Alidra Mohamed Ouassim

Backend Developer

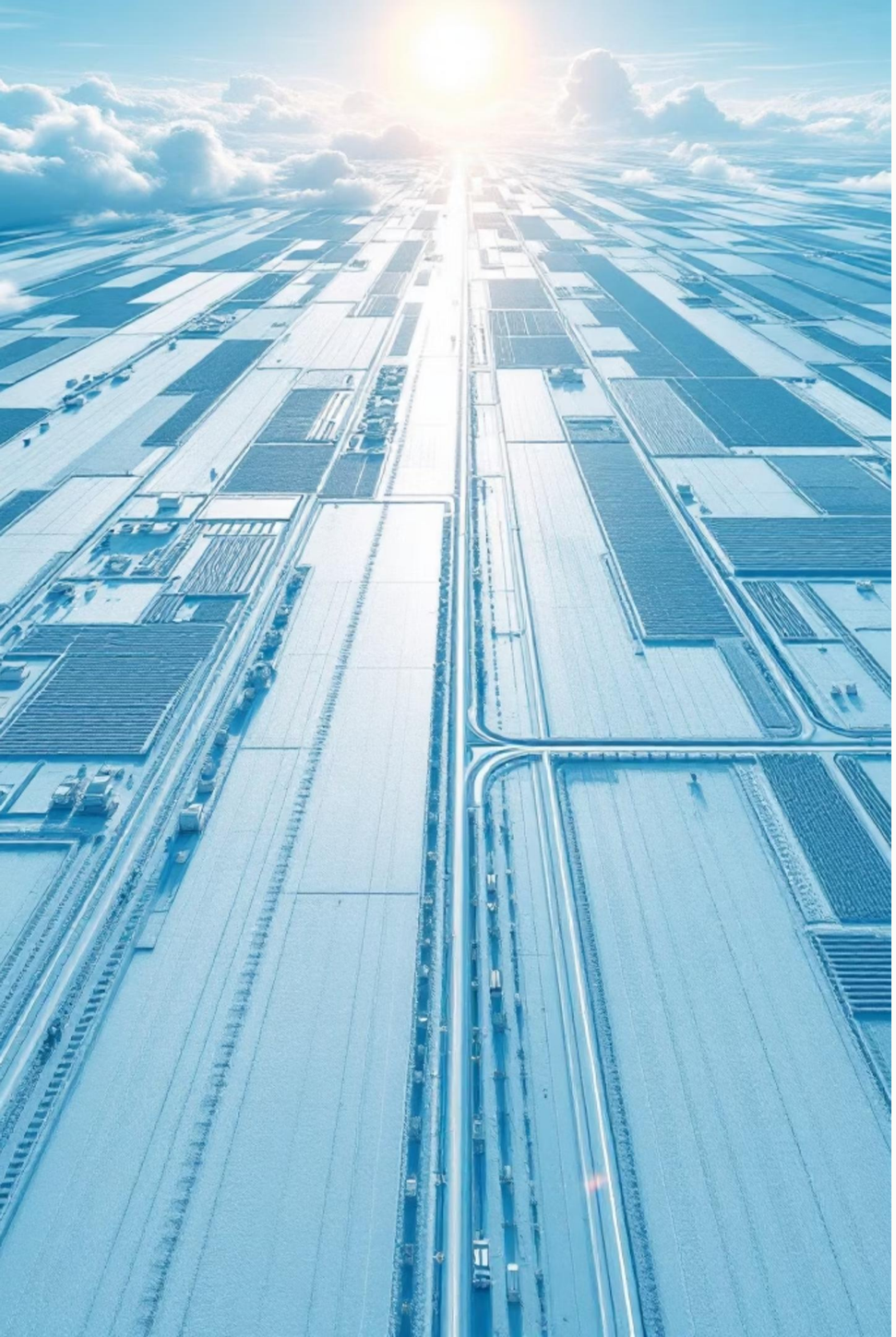
Chedadi Ahmed Anis

UX/UI Designer

Benouarzeg Mohamed Ali

Project Manager / Tester

Benmamar Faiz El Maamoune



Problem Context

Agriculture is a key sector in Algeria, playing a vital role in the national economy and food security.

However, many farmers rely on intuition instead of data-driven insights for their operations. This leads to several inefficiencies:

Water waste

Inefficient irrigation practices due to lack of precise data.

Low crop yield

Suboptimal growing conditions and resource management.

Poor decision-making

Choices based on guesswork rather than actionable intelligence.

Existing smart farming solutions often fail to address these issues effectively because they are:

Expensive

Prohibitive costs for Algerian farmers to adopt.

Not adapted to local needs

Lacking customization for Algeria's specific climate, crops, and agricultural practices.

Main Challenge: How can we help Algerian farmers make smarter and more sustainable decisions?

Brainstorming Methodology

We used multiple ideation techniques:

Traditional Brainstorming

Brainwriting

Structured Evaluation

Idea Evaluation

Idea	Feasibility	Cost	Impact
Mobile App Only	High	Low	Medium
Drone Monitoring	Medium	High	High
AI Disease Detection	Medium	Medium	High
Smart Irrigation System	High	Medium	Very High
IoT + Mobile App	High	Medium	High
Integrated Platform	High	Medium	Very High

Selected Solution: Integrated precision farming platform combining IoT, AI, weather support, and irrigation management.

Proposed Solution: AgroSmart DZ

A smart farming platform that includes:

IoT Sensors

- Soil moisture
- Temperature
- Humidity

Mobile Application

- Displays real-time data
- Sends alerts and recommendations

AI Engine

- Irrigation recommendations
- Risk warnings

Weather Integration

- Forecast support
- Climate-based decisions

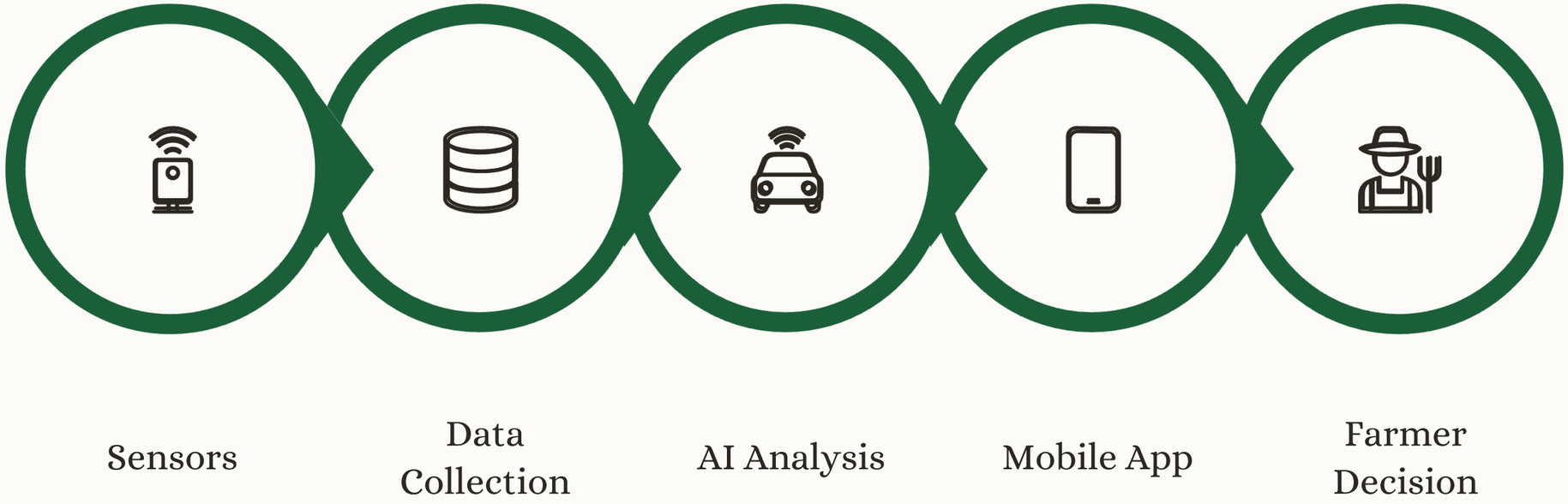
Smart Irrigation Support

- Optimized water usage

Goal:

Move from intuition-based farming to data-driven farming

System Workflow



Sensors collect real-time field data.

AI analyzes the data.

App sends alerts and recommendations.

Farmer takes better decisions.

Usage Scenario

Before AgroSmart DZ

- Irrigation based on habit
- Water waste
- Unstable productivity

After AgroSmart DZ

- Real-time soil monitoring
- Smart irrigation recommendations
- Water saving
- Improved crop yield

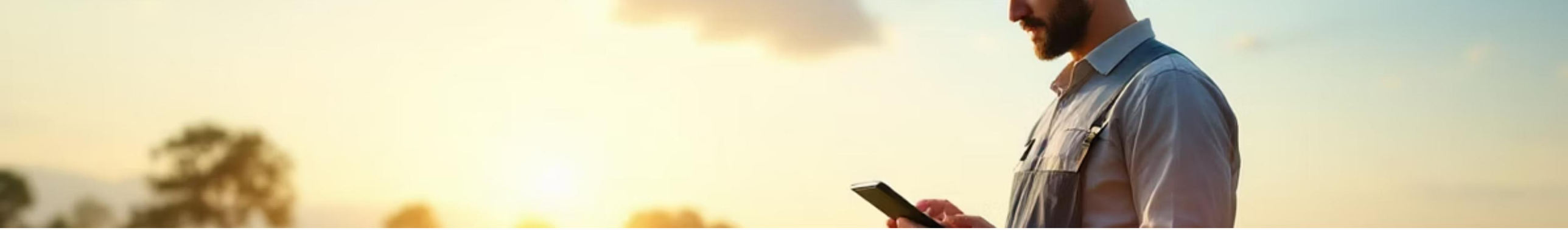
Key Features

Current Features

- Mobile farming application
- Smart irrigation system
- IoT soil monitoring
- Weather assistant

Future Features

- Drone monitoring
- AI disease detection



Why Invest in AgroSmart DZ?

- Solves a real local problem in Algeria
- Supports sustainable agriculture
- Optimizes water consumption
- Strong social and environmental impact
- Scalable to multiple farms and regions

Value Proposition:

Affordable, practical, and impactful smart farming platform

Future Implementation

Planned enhancements:

1. Drone-based crop monitoring
2. AI disease detection using images
3. Market price prediction
4. Multilingual support
5. Integration with agricultural institutions

Conclusion

Key Takeaways

- A structured brainstorming process was used
- AgroSmart DZ is the selected solution
- Combines IoT + AI for smarter agriculture

Benefits

1. Farmers
2. Environment
3. Agricultural sector

Thank You

Questions?

We appreciate your time and interest.